Part I Introductory Topics For Everyone

1) Introduction and Motivation

1) Online Controlled Experiments Terminology.

#Overview

"online Controlled Experiments Terminology" introduces standard vocabulary used throughout the book and in the practice of A/B testing (online experiments). Clear understanding of these terms is crucial for designing, analyzing and interpreting experiments.

☑ Core terminology (Explained Simply with Analogies + Math + Code).

1. Experiment (A/B Tests)

Definitions: A scientific tests where user are randomly split into two or more groups.

- control Group (A): Gets the original experience.
- Treatment Groups (B): Gets the modified/new experience

Analogy: like testing two types of fertilizers on similar plots of lands. Group A gets old fertilizers; group B gets new fertilizers. We compare plant growth

Math Concept: Random assignment ensures independent samples so we can validly compare means/metrics.

2. Treatment Effect

Definition: The difference in outcomes between control and treatment. This is the effect caused by the change.

Formula: Treatment Effect = $\bar{x}_{treatment} - \bar{x}_{control}$.

3. Randomization

Definition: Randomly assigning users to control or treatment groups to eliminate bias.

Analogy: Tossing a fair coin to assign users — no favoritism or skewed distribution.

Math Concept: Random assignment → balanced groups
↳ valid inference via CLT

4. Unit of Randomization / Analysis

Definition: The entity over which randomization is done and analysis is measured.

Common Units:
- user (most common)
- session (can bias metrics)
- Cookie / device (less stable).

Interlinked Concept: Bias and confounding if wrong unit is used.

Example: If a user clears cookies and is treated again, it breaks independence assumption.

5. Metric

Definition: A quantifiable measure used to compare performance between control and treatment.

Types:

- Invariant Metric: should not change (e.g. user count)
- Evaluation Metric: Measure success (e.g clicks, revenue)

Analogy: In a school exam experiment, invariant metric = number of students, evaluation metric = average score.

6) Null Hypothesis & Statistical Significance

Null Hypothesis (H_0): No difference between control and treatment.

Statistical Significance: How confident we are that the observed effect isn't due to chance.

Math Formula:

$$Z = \frac{\bar{X}_T - \bar{X}_C}{\sqrt{\frac{\sigma_T^2}{n_T} + \frac{\sigma_C^2}{n_C}}}$$

7) Power and Sample Size.

- Power: Probability that your experiment will detect a true effect.
- Same Size: Need to ensure experiment has sufficient power.

8. Lift

Definition: Percent increase (or decrease) in metric compared to control.

Formula: $\text{Lift} = \frac{\bar{X}_T - \bar{X}_c}{\bar{X}_c} \times 100\%$

9. Variance & Standard Error.

Variance: Spread of the data

Standard Error: Variability of the mean across samples.

Used for: Confidence intervals and hypothesis testing.

10. Interlocked Topics.

These terms are tied tightly with:

- Why Experiment? - Because of causality vs correlation.

10. P-value Misterpretation

Clarification: A low p-value doesn't prove the treatment is good - just that it is unlikely to observe the result by chance under the null.

② Why Experiments? Correlations, Causality, and Trustworthiness &

Goal of This Section.

To answer one big question:

| "Why do we need to run online experiment (like A/B tests) instead of just looking at data?"

1. Correlation: When Two Things Happen Together

Definition

| Correlation means two thing change together - but we don't know why.

Example

Let's say you see this in your app data:

- Users who spend more time on the site also buy more.

This is a correlation - two things go up together

But you don't know if:

- A: Time causes purchases
- B: Purchases cause time
- C: A third reason (like promotions) causes both.

2. Causation: When One Thing Causes Another.

Definition

| causation means: "changing X directly change Y"

You need intervention - change something and measure the effect.

Real-world Analogy

Imagine:

- You see that students who take online notes score higher.
- So you say: "Taking online notes causes better grades."

But what if:

- These students were already smarter?
- Or more motivated?

⇒ You don't trust that correlation. You must test it.

☑ Controlled experiment

- Randomly assign students to "take notes online" vs "papers only"
- Then compare their scores.

Now, you know whether the method causes better results.

3. Why Experiment?

| Because randomized experiment are the only trustworthy way to measure causality

☑ What Experiments Do:

1. Randomly assign users to control and treatment groups.
2. Apply your change (like a new button) only to treatment
3. Measure results.

If the treatment group performs better, you can say:
⇒ "This change caused the improvement".

Trustworthiness.

- No bias: Random assignment makes both group equal on average.
- Fairness: You isolate only the change, not other effects.
- Reliable decisions: You can act based on real evidence.

(3) Necessary Ingredients for Running Useful Controlled Experiments

The authors stress that just running an experiment isn't enough — to get useful, trustworthy results you need certain core ingredients.

1. Clear Hypothesis
2. Randomization.
3. Sufficient sample size (Statistical Power)
4. Reliable Instrumentation (Accurate Data Collection).
5. Relevant Metrics.
6. Test Duration (Run the test long enough to cover variations (day of week, seasonality, user habits).
7. Analysis plan

Decide beforehand how you'll analyze results, including:

- Hypothesis test type
- Significance threshold
- Segments to analyze.

Why?
Avoids cherry-picking results after the fact.

4. Tenets

The authors introduce a guiding principle ("Tenets") that every online experiments should follow. These are like rules of the road for A/B testing – they keep answers trustworthy, reproducible and actionable.

1. Trust the data - But Ouery Tools Properly Run Experiments
2. Speed Matters - Run Fast - Learn Fast.
3. Decentralize - let Teams Test Independently.
4. Evaluate long-term Impact, Not just short-term Goal.
5. Don't Overreact to Noisy Data.
6) Make Decisions Based on Pre-Defined Metrics.
7. Beware of Launching without Testing.
8. Understand Practical vs Statistical Significance.
9. Don't Stop Tests Early Just Because They look Good.

5. Improvement over Time

(1) Small changes. Big Gain Over Time

(2) Compound Effect of Iteration.

If each experiment increases a metric by even 1%, repeated over multiple cycles, the cumulative effect is substantial.

(3) Speed and Feedback loop.

- Insight: Rapid experimentation accelerates Learning. Shorter cycle times lead to more tests – and more opportunities for improvement.

- connections:
- ~~Insight~~:
- Tactics: "speed Matters" encourages just this iterative flow.
- Necessary: Requires good instrumentation in products, metrics, and automation to support rapid.

(4) Avoiding Over-Reliance on Big Bets (Changes).

(5) Examples of Interesting Online Controlled Experiments is

(1) What The's unit Covers

→ strategy refers to the overarching goal and direction- what you ultimately want to achieve.

→ Tactics are the specific actions or experiments you perform to move toward that strategic goal.

(2) Strategy vs. Tactics: Definitions & Analogy.

- Strategy
  - Big-picture aims (e.g. "Increase coverage order value by 10% this quarter")
  - Determines which metrics matter most.
- Tactics
  - Concrete experiments (e.g. "Test a free-shipping banner at checkout")
  - Measured via A/B tests to check their contribution toward strategic goals.

Real-world analogy

Strategy is the destination ("Climb Mount Everest")
: tactics are the steps (pick route, pack gear, train).
Experiments help test each tactic's effectiveness before committing.

Math Concept: Strategy focuses on OEC, Tactics Deliver lift.

In the book, authors recommend defining an Overall Evaluation Criterion (OEC) - a single metric tied to strategy - and the running multiple experiments (tactics) that together optimize it.

If each tactic yields a small attributable lift Δ_i, over n tactics

$$\text{Total lift} \approx \sum_{i=1}^{n} \Delta_i$$

Experiment quantifies each Δ_i, so strategy is evidence-driven.

(7) Additional Reading 24 ✓.